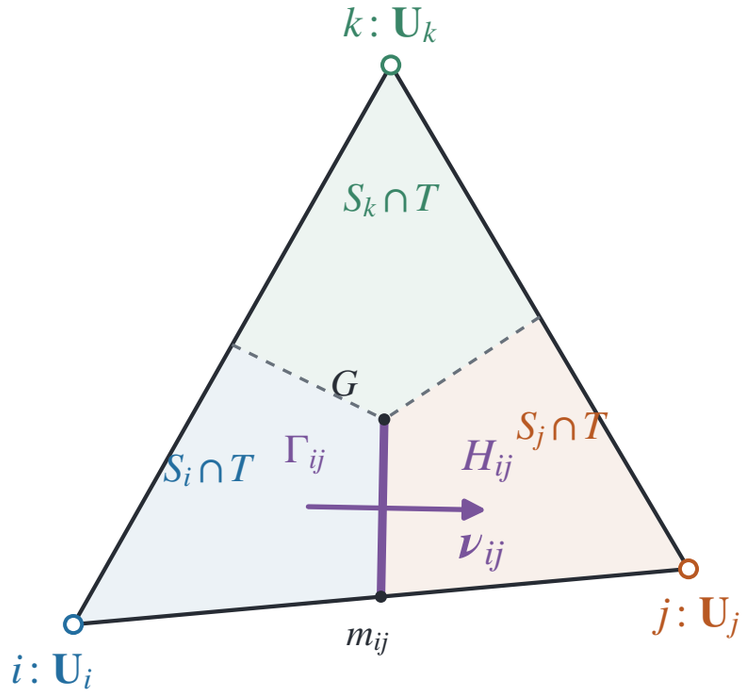
From a Riemann flux to residual distribution

(a) Evaluate the numerical flux

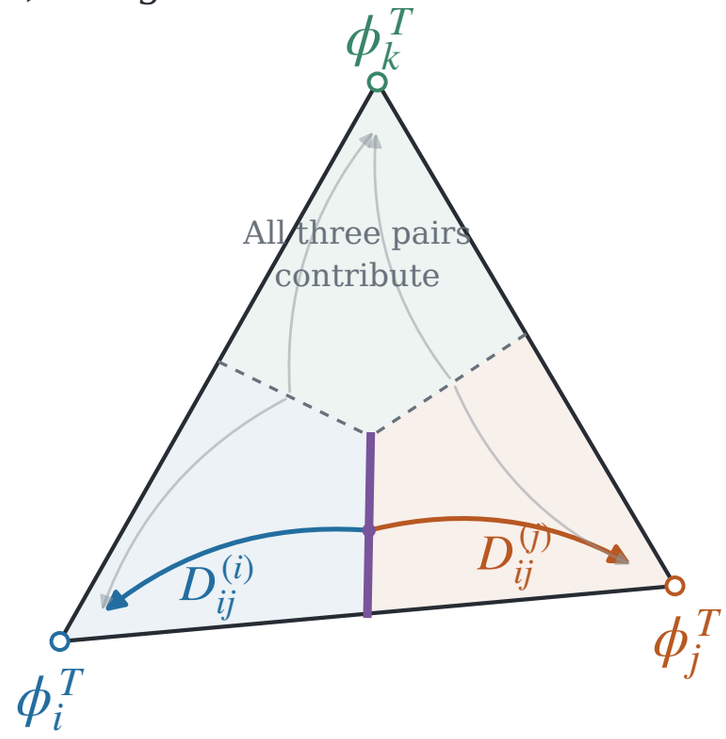


$$H_{ij} = |\Gamma_{ij}| \widehat{f}_{\hat{\mathbf{n}}_{ij}}(\mathbf{U}_i, \mathbf{U}_j)$$

$$\boldsymbol{\nu}_{ij} = |\Gamma_{ij}| \hat{\mathbf{n}}_{ij} = (\mathbf{n}_j - \mathbf{n}_i)/6$$

One normal Riemann problem: HLLC, HLL, Roe, ...

(b) Assign fluctuations to the vertices



$$D_{ij}^{(i)} = H_{ij} - \mathbf{F}_i \cdot \boldsymbol{\nu}_{ij}$$

$$D_{ij}^{(j)} = \mathbf{F}_j \cdot \boldsymbol{\nu}_{ij} - H_{ij}$$

$$\phi_i^T = \sum_{j \neq i} D_{ij}^{(i)}$$

$$D_{ij}^{(i)} + D_{ij}^{(j)} = (\mathbf{F}_j - \mathbf{F}_i) \cdot \boldsymbol{\nu}_{ij}$$

$$\Longrightarrow \qquad \phi_i^T + \phi_j^T + \phi_k^T = \frac{1}{2} \sum_{a \in T} \mathbf{F}_a \cdot \mathbf{n}_a = \Phi^T$$

Pairwise numerical fluxes cancel; the element residual is conserved.

$\mathbf{F}_a = \mathbf{F}(\mathbf{U}_a)$; $\mathbf{n}_a$: inward normal to the opposite edge, scaled by edge length.

Static spatial construction with interpolated nodal fluxes. Arrows in (b) denote assignment, not fluctuation signs.